Decomposition (the alternative of prime's powers in a common ring)

definitions and properties

VI.1.1. $\mathfrak{q} \subsetneq A$ is an ideal, then $\mathfrak{q}$ is primary iff $xy \in \mathfrak{q} \Rightarrow x \in \mathfrak{q}$ or $y^n \in \mathfrak{q}$

Remark $\mathfrak{q}$ is primary $\Leftrightarrow A/\mathfrak{q} \neq 0$, any zero-divisor in $A/\mathfrak{q}$ is nilpotent

Prop. VI.1.2 $\mathfrak{q}$ is a primary ideal in A, then $\sqrt{\mathfrak{q}}$ is the smallest prime ideal containing $\mathfrak{q}$

pf: By the previous proposition, $\sqrt{\mathfrak{q}} = \bigcap_{\mathfrak{q} \subseteq \mathfrak{p}_i} \mathfrak{p}_i := \mathfrak{p} \supseteq \mathfrak{q}$

$\forall xy \in \mathfrak{p} \Rightarrow (xy)^m \in \mathfrak{q} \Rightarrow x \in \sqrt{\mathfrak{q}}$ or $\sqrt{\mathfrak{q}}$ thus $\mathfrak{p}$ is a prime ideal ▨

Remark $\mathfrak{q}$ is said to be $\mathfrak{p}$-primary, if $\mathfrak{q}$ is primary and $\sqrt{\mathfrak{q}} = \mathfrak{p}$

Examples. 1) The primary ideals in $\mathbb{Z}$ are (0) and (p^n) where p is prime,

2) $A = k[x,y]$, $\mathfrak{q} = \langle x, y^2 \rangle$ Then $\sqrt{\mathfrak{q}} = \langle x, y \rangle$, $(A/\mathfrak{q} \cong k[y]/(y^2) \Rightarrow \mathfrak{q}$ is primary.
$\mathfrak{p}^2 \subsetneq \mathfrak{q} \subsetneq \mathfrak{p}$

Prop. VI.1.3 If $\sqrt{\mathfrak{a}}$ is maximal, then $\mathfrak{a}$ is primary. The powers of a maximal ideal $\mathfrak{m}$ is $\mathfrak{m}$-primary.

pf: let $\sqrt{\mathfrak{a}} = \mathfrak{m}$, then $\sqrt{A/\mathfrak{a}} = \operatorname{nil}(A/\mathfrak{a}) = \pi(\sqrt{\mathfrak{a}})$ is maximal

$\Rightarrow A/\mathfrak{a}$ has only one prime ideal $\Rightarrow$ every zero-divisor in nilpotent in $A/\mathfrak{a}$ ▨

Lemma VI.1.4. $\mathfrak{q}_i$ $(1 \le i \le n)$ are $\mathfrak{p}$-primary, then $\mathfrak{q} = \bigcap_{i=1}^{n} \mathfrak{q}_i$ are $\mathfrak{p}$-primary

pf: $\sqrt{\mathfrak{q}} = \sqrt{\bigcap_{i=1}^{n} \mathfrak{q}_i} = \bigcap_{i=1}^{n} \sqrt{\mathfrak{q}_i} = \mathfrak{p}$; $\forall xy \in \mathfrak{q}$, $y \notin \mathfrak{q} \Rightarrow \exists i$ s.t. $xy \in \mathfrak{q}_i$, $y \notin \mathfrak{q}_i$

$\Rightarrow x \in \mathfrak{p}$. ▨

Lemma VI.1.5. $\mathfrak{q}$ be a $\mathfrak{p}$-primary ideal $x \in A$ Then

i). if $x \in \mathfrak{q}$ then $(\mathfrak{q} : x) = (1)$

ii) if $x \notin \mathfrak{q}$ then $(\mathfrak{q} : x)$ is $\mathfrak{p}$-primary, therefore $\sqrt{(\mathfrak{q} : x)} = \mathfrak{p}$

iii) if $x \notin \mathfrak{p}$ then $(\mathfrak{q} : x) = \mathfrak{q}$

pf: i) & iii) is obvious.

ii) $y \in (\mathfrak{q} : x) \Leftrightarrow xy \in \mathfrak{q} \Rightarrow y \in \mathfrak{p} \Rightarrow \mathfrak{q} \subseteq (\mathfrak{q} : x) \subseteq \mathfrak{p} \Rightarrow \sqrt{(\mathfrak{q} : x)} = \mathfrak{p}$.

$yz \in (\mathfrak{q} : x)$, $y \notin \mathfrak{p}$ $\Rightarrow xyz \in \mathfrak{q} \Rightarrow xz \in \mathfrak{q} \Rightarrow z \in (\mathfrak{q} : x)$ ▨

Def VI 1.5 $\mathfrak{q} \subseteq A$ Suppose that $\mathfrak{q}$ has a prim

$$\mathfrak{q} = \bigcap_{i=1}^{n} \mathfrak{q}_i \quad \sqrt{\mathfrak{q}_i} \neq \sqrt{\mathfrak{q}_j}, \forall i \neq j \quad \text{and} \quad \mathfrak{q}_j \not\supseteq \bigcap_{i \neq j} \mathfrak{q}_i$$

Then we call $\mathfrak{q}$ is decomposable

Thm VI 1.6 (Uniqueness thm) $\mathfrak{q} \subseteq A$ is ~~dege~~ decomposable, $\mathfrak{q} = \bigcap_{i=1}^{n} \mathfrak{q}_i$ minimal, $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$

Then $\{\mathfrak{p}_i \mid 1 \le i \le n\} = \{\mathfrak{p} \mid \mathfrak{p} = \sqrt{(\mathfrak{q} : x)},\ x \in A\}$ (*)

pf: $(\mathfrak{q} : x) = \bigcap_{i=1}^{n} (\mathfrak{q}_i : x) \Rightarrow \sqrt{(\mathfrak{q} : x)} = \bigcap_{i=1}^{n} \sqrt{(\mathfrak{q}_i : x)} \overset{(1.5)}{=\!=} \bigcap_{x \notin \mathfrak{q}_j} \mathfrak{p}_j$

① $\sqrt{(\mathfrak{q} : x)}$ is prime $\Rightarrow \exists j$ s.t. $\sqrt{\mathfrak{q} : x} = \mathfrak{p}_j$ (Prime avoidance).

② $\forall i \in [1,n]$, $\exists x_i$ s.t. $x_i \notin \mathfrak{q}_i$ ●, $x_i \in \bigcap_{j \neq i} \mathfrak{q}_j \Rightarrow \sqrt{(\mathfrak{q} : x_i)} = \mathfrak{p}_i$ ▨

Remark. 1) $(\mathfrak{q} : x_i) = (\mathfrak{q}_i : x)$ is $\mathfrak{p}_i$ - primary.

2). $A/\mathfrak{q}$ is an A-module, then $(\mathfrak{q} : x_i) = \mathrm{Ann}_{A/\mathfrak{q}}(x_i)$

3). We call $\mathfrak{p}_i$ in (*) is associated with $\mathfrak{q}$, the minimal elements in $\{\mathfrak{p}_i \mid 1 \le i \le n\}$. is called the minimal/isolated prime ideals belonging to $\mathfrak{q}$, the others are called embedded prime ideals

Examples: 1) $\mathfrak{q} = (x^2, xy)$. $A = k[x,y]$ Then $\mathfrak{q} = \mathfrak{p}_1 \cap \mathfrak{p}_2^2$ where $\mathfrak{p}_1 = (x)$, $\mathfrak{p}_2 = (x,y)$.

Then $\mathfrak{p}_1 \neq \mathfrak{p}_2$ &, $\mathfrak{p}_1$ is the isolated ideals. $\mathfrak{p}_2$ is the embedded ideal.

2).

Pp. VI.1.7. $\mathfrak{q}$ decomposable, then $\{\mathfrak{p} \mid \mathfrak{p} \supseteq \mathfrak{q}, \mathfrak{p}$ minimal$\}$ $\overset{A}{=\!=}$ $\{\mathfrak{p} \mid \mathfrak{p}$ is the minimal/isolated ideal belonging to $\mathfrak{q}\}$ $\overset{B}{=\!=}$

pf: ① $\forall \mathfrak{p} \in A$. $\Rightarrow \mathfrak{p} \supseteq \bigcap_{i=1}^{n} \mathfrak{q}_i \Rightarrow \mathfrak{p} = \sqrt{\mathfrak{p}} \supseteq \bigcap_{i=1}^{n} \mathfrak{p}_i$

By prime avoidance lemma, $\exists i$, s.t. $\mathfrak{p} \supseteq \mathfrak{p}_i$ but $\mathfrak{p}_i \supseteq \mathfrak{q}_i \supseteq \mathfrak{q}$

By the definition of A, we have $\mathfrak{p} = \mathfrak{p}_i \in B$

② $\forall \mathfrak{p}_i \in B$. $\mathfrak{p}_i \supseteq \mathfrak{q}$ if $\exists \mathfrak{p} \in A$ s.t. $\mathfrak{p}_i \supseteq \mathfrak{p}$. then by ①. $\exists j$. s.t. $\mathfrak{p} = \mathfrak{p}_j \in B$.

By the definition of B. $\mathfrak{p}_i = \mathfrak{p} = \mathfrak{p}_j$ ▨

Pp. VI.1.8. $\mathfrak{q}$ decomposable, $\mathfrak{q} = \bigcap_{i=1}^{n} \mathfrak{q}_i$ minimal, $\sqrt{\mathfrak{q}_i} = \mathfrak{p}_i$ Then $\bigcup_{i=1}^{n} \mathfrak{p}_i = \{x \in A : (\mathfrak{q} : x) \neq \mathfrak{q}\}$.

pf: In $A/\mathfrak{q}$, $(0) = \bigcap_{i=1}^{n} \overline{\mathfrak{q}_i} \Rightarrow \overline{D} = \bigcup_{\overline{y} \neq 0} (\overline{0} : \overline{y}) = \bigcup_{\overline{y} \neq 0} \sqrt{\overline{0} : \overline{y}} = \bigcup_{\overline{y} \neq 0} \bigcap_{\overline{y} \notin \overline{\mathfrak{q}_j}} \overline{\mathfrak{p}_j} \subseteq \bigcup_{i=1}^{n} \overline{\mathfrak{p}_i}$

By. (1.6). we have $\bigcup_{i=1}^{n} \overline{\mathfrak{p}_i} \subseteq \overline{D} \Rightarrow \overline{D} = \bigcup_{i=1}^{n} \overline{\mathfrak{p}_i}$.

Taking π^{-1}: $\bigcup_{i=1}^{n} \mathfrak{p}_i = \{x \in A : (\mathfrak{q} : x) \neq \mathfrak{q}\}$. ▨

$P = \bigcup$ all prime ideals belonging to 0;

$R = \mathrm{nil}(A) = \bigcap$ all minimal prime ideals belonging to 0

(If the zero ideal is decomposable)

Prop. VI.1.9 (Primary ideals under localization). S is a multiplicatively closed subset of A.

$\mathfrak{q}$ be a $\mathfrak{p}$-primary ideal.

Then i). $S \cap \mathfrak{p} \neq \phi \Rightarrow S^{-1}\mathfrak{q} = S^{-1}A$

ii) $S \cap \mathfrak{p} = \phi \Rightarrow S^{-1}\mathfrak{q}$ is $S^{-1}\mathfrak{p}$-primary. and $(S^{-1}\mathfrak{q})^c = \mathfrak{q}$.

pf: i) is obvious.

ii). $S \cap \mathfrak{p} = \phi \Rightarrow \forall s \in S.\ x \in A,\ sx \in \mathfrak{q} \Rightarrow x \in \mathfrak{q}$ $\Rightarrow$ let $x/1 \in S^{-1}\mathfrak{q} \Rightarrow \exists t \in S$ s.t. $tx \in \mathfrak{q} \Rightarrow x \in \mathfrak{q}$

$\Rightarrow \mathfrak{q}^{ec} = \mathfrak{q}$, Also $\sqrt{S^{-1}\mathfrak{q}} = S^{-1}\mathfrak{p}$. $\Rightarrow S^{-1}\mathfrak{q}$ is $S^{-1}\mathfrak{p}$-primary ▨

Notation: let $S(\mathfrak{a})$ be the contraction of $S^{-1}\mathfrak{a}$

Prop. VI.1.10. let $\mathfrak{a} = \bigcap_{i=1}^{n} \mathfrak{q}_i$ minimal, $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$. And $\mathfrak{p}_i \cap S = \phi$ $(1 \le i \le m)$ $\mathfrak{p}_i \cap S \neq \phi$ $(i \ge m+1)$

Then $S^{-1}\mathfrak{a} = \bigcap_{i=1}^{m} S^{-1}\mathfrak{q}_i$. $S(\mathfrak{a}) = \bigcap_{i=1}^{m} \mathfrak{q}_i$.

pf: (skipped).

Remark. A set Σ of prime ideals belonging to $\mathfrak{a}$ is isolated iff.: if $\mathfrak{p}'$ belongs to $\mathfrak{a}$ and $\mathfrak{p}' \subseteq \mathfrak{p}$ for some $\mathfrak{p} \in \Sigma$. then $\mathfrak{p}' \in \Sigma$.

Let $S = A - \bigcup_{\mathfrak{p} \in \Sigma} \mathfrak{p}$, then S is a multiplicatively ~~isolated~~ closed set.

$\mathfrak{p}' \in \Sigma \Rightarrow \mathfrak{p}' \cap S = \phi$

$\mathfrak{p}' \notin \Sigma \Rightarrow \mathfrak{p}' \nsubseteq \bigcup_{\mathfrak{p} \in \Sigma} \mathfrak{p}$ (Prime avoidance) $\Rightarrow \mathfrak{p}' \cap S \neq \phi$

By (VI.1.10) We have:

Thm VI.1.11. (Uniqueness thm) $\mathfrak{a}$ decomposable. $\mathfrak{a} = \bigcap_{i=1}^{n} \mathfrak{q}_i$ minimal, Σ is an isolated set of prime ideals of $\mathfrak{a}$. Then $\bigcap_{\mathfrak{p} \in \Sigma} \mathfrak{q}$ is independent of decomposition.

pf: (skipped).

Cora VI.1.12. The isolated primary components are uniquely determined by $\mathfrak{a}$.

pf: $\forall\ \mathfrak{q}_i$ corresponds to minimal primes ideals $\mathfrak{p}_i$. Take $\Sigma = \{\mathfrak{p}_i\}$ ▨